Budget & Expense Analytics

Final Analytical Report — Demonstration Version

Critical limitation: This package uses synthetic demonstration data because no company dataset was provided. The figures and findings below only demonstrate that the analysis pipeline works; they are not real company findings and must not be submitted as evidence about an actual organization. Replace the CSVs with verified company records and regenerate this report.

1. Problem and objectives

The project demonstrates how to combine approved budget and actual expense records by month, department, and expense category; calculate budget variance and utilization; detect over-budget combinations; and present results through a filterable Streamlit dashboard.

2. Data and preparation

The demo contains 360 budget rows and 360 expense rows for 12 months, five departments, and six categories. All values are marked synthetic and use PKR only as a demonstration currency label. The workflow parses month values, trims text keys, converts financial fields to numeric, drops invalid required fields, aggregates duplicate keys, and performs a validated one-to-one merge.

Dataset source: no real company data was supplied. The included records were generated for software testing and are not an external dataset.

3. KPI definitions

• Approved budget: sum of approved allocations for the selected period and filters.

• Actual expenses: sum of recorded expenses for the same period and filters.

• Variance: approved budget minus actual expenses; positive values mean under budget and negative values mean overspending.

• Budget utilization: actual expenses divided by approved budget, multiplied by 100; only meaningful where budget is greater than zero.

• Over-budget alert: actual expense is greater than approved budget for a month/department/category combination.

4. Demo output (synthetic data only)

Demo approved budget: PKR 10,145,340. Demo actual expense: PKR 9,637,198. Demo variance: PKR 508,142. Demo utilization: 95.0%. Demo over-budget combinations: 67. These values describe only the generated sample.

5. Example observations — not real business findings

• In the generated sample, IT has the lowest department-level variance (PKR 55,593); this is a deliberate test pattern, not a claim about a real company.

• The generated sample's category with the lowest variance is Software & IT (PKR 26,664); validate category mappings and actual invoices in real data before drawing conclusions.

• The month with the lowest overall variance in the generated sample is 2025-07 (PKR 8,361).

• The sample includes 67 month/department/category combinations where actual expense exceeds budget, allowing the dashboard alert table to be tested.

• The data includes test patterns for higher IT software costs in the second half of the year, seasonal utility expenses, and increased marketing travel in selected months; these are intentionally programmed into the sample and are not discovered real-world effects.

6. Actionable cost-control recommendations

• Use monthly budget-owner reviews: require a brief explanation and corrective action for every verified combination with utilization above 100%.

• Set early-warning thresholds (for example, 85% utilization before the final month) and notify department owners when actuals plus committed spend forecast a breach.

• Improve procurement controls: compare recurring software subscriptions, travel requests, and supplier rates against approved plans; require approval for out-of-budget commitments.

• Standardize department and category codes in the source systems so that monthly merges do not split the same cost across inconsistent labels.

• Reconcile dashboard totals to finance-system totals each month and retain the source file, extraction date, and cleaning log for auditability.

7. Dashboard and deliverables

The Streamlit dashboard includes date, department, and category filters; four financial KPI cards; monthly budget-versus-actual trend; department comparison; category mix; monthly variance; an over-budget alert table; detailed records; and filtered CSV download. The Jupyter notebook documents cleaning, joining, calculations, charts, and exports.

8. Limitations and next steps

The sample is not representative of any company, contains no invoices or committed-spend pipeline, and uses illustrative allocations. Before final submission, replace the CSVs with authorized company data, document the source, check for missing/unmatched keys and outliers, rerun the notebook, and replace demo-only findings with at least five findings calculated from real records.

Appendix: Data source and reproducibility

See README.md for setup, run commands, expected column names, KPI definitions, and the required real-data validation steps.